

Solution key and worked guide

Six worked solutions, with the method behind each answer. Based on the supplied assignment due October 7, 2026. This is an independently prepared study key, not an instructor-issued answer key.

Quick answer key

Question	Answer
1. MgO density	3.600 g/cm³
2. One mole of Mg	2.41 cm × 2.41 cm × 2.41 cm Using Mg density = 1.74 g/cm ³ .
3. Cubic directions	35.26° between [110] and [111].
4. Plane (3, -1, 1)	x = a/3, y = -b, z = c Normalized intercepts: (1/3, -1, 1).
5. Vacancy fraction	8.81 × 10⁻⁴ , or 0.0881% Using the supplied fraction at 400°C.
6. Carburizing time	3.69 × 10⁴ s ≈ 10.2 h Small variation comes from error-function table rounding.

How to present your work

For each question: list the given quantities, write the governing relation, substitute values with units, and box the final answer. Keep extra digits until the final step. For Question 6, show the table lookup or interpolation explicitly.

Avogadro's number is supplied on the assignment, but none of these six solutions needs it: the first two questions already specify the amount in moles.

Mass, volume, and cube size

1 / Density of solid MgO

Given: one mole of MgO occupies a cube with side $L = 22.37$ mm. Find density in g/cm^3 .

Step 1 - Find the mass of one mole. Each MgO formula unit contains one Mg and one O. Using molar masses $M_{\text{g}} = 24.305$ and $O = 15.999$ g/mol:

$$M_{\text{MgO}} = 24.305 + 15.999 = 40.304 \text{ g/mol}$$

$$m = nM = (1 \text{ mol})(40.304 \text{ g/mol}) = 40.304 \text{ g}$$

Step 2 - Convert the side length, then cube it.

$$L = 22.37 \text{ mm} = 2.237 \text{ cm}$$

$$V = L^3 = (2.237 \text{ cm})^3 = 11.1943 \text{ cm}^3$$

Step 3 - Divide mass by volume.

$$\text{Density} = m/V = 40.304 / 11.1943$$

$$\text{Density} = 3.600 \text{ g/cm}^3$$

Common mistake: converting mm^3 as though it were mm. Since $1 \text{ cm} = 10 \text{ mm}$, $1 \text{ cm}^3 = 1000 \text{ mm}^3$. Converting the length first avoids this error.

2 / Dimensions of a cube containing one mole of Mg

Needed reference value: the assignment does not supply magnesium density. Use the standard room-temperature value 1.74 g/cm^3 and molar mass 24.305 g/mol . If your course table specifies another density, substitute it in the same formula.

$$m = nM = 24.305 \text{ g}$$

$$V = m / \text{density} = 24.305 / 1.74 = 13.9684 \text{ cm}^3$$

$$L = V^{1/3} = (13.9684)^{1/3} = 2.40833 \text{ cm}$$

$$\text{Dimensions} = 2.41 \text{ cm} \times 2.41 \text{ cm} \times 2.41 \text{ cm}$$

Check: Mg has less mass per mole than MgO but much lower density, so its one-mole cube is larger: about 24.1 mm per side versus 22.37 mm. This is a bulk sample, not a unit cell.

Mg reference: [Royal Society of Chemistry, Magnesium](#) (density 1.74 g/cm^3 ; relative atomic mass 24.305).

Directions and plane intercepts

3 / Angle between [110] and [111]

In a cubic system, the three axes are perpendicular and have equal scale. The indices can therefore be used as Cartesian vector components: $\mathbf{u} = (1, 1, 0)$ and $\mathbf{v} = (1, 1, 1)$.

$$\cos(\theta) = (\mathbf{u} \cdot \mathbf{v}) / (|\mathbf{u}||\mathbf{v}|)$$

$$\mathbf{u} \cdot \mathbf{v} = (1)(1) + (1)(1) + (0)(1) = 2$$

$$|\mathbf{u}| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

$$|\mathbf{v}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos(\theta) = 2 / \sqrt{6} = 0.816497$$

$$\theta = \arccos(0.816497) = 35.26^\circ$$

Check: [110] lies along a face diagonal, while [111] points along the body diagonal. The angle between these specified directions is acute. Set the calculator to degrees.

4 / Axial intercepts of the (3, -1, 1) plane

In the original document, the bar is over the **middle 1**. Thus $h = 3$, $k = -1$, and $l = 1$. Commas here only make the three indices easier to distinguish.

Miller indices are reciprocals of the intercepts measured in lattice units. Reverse that operation to find the intercepts:

$$x/a = 1/h = 1/3$$

$$y/b = 1/k = -1$$

$$z/c = 1/l = 1$$

Axial intercepts: $a/3$, $-b$, c

The actual points on the three axes are $(a/3, 0, 0)$, $(0, -b, 0)$, and $(0, 0, c)$. For a cubic lattice, $a = b = c$, giving $a/3$, $-a$, a .

How to sketch and verify the plane

Draw the x , y , and z axes, including the negative y direction. Mark the three intercept points and connect them to form a triangular portion of the plane. The negative intercept means the plane crosses the negative y axis.

$$\text{Plane equation: } 3x/a - y/b + z/c = 1$$

Substituting any one of the three intercept points gives 1. Do not multiply the intercepts by 3 just to remove the fraction: that selects a different parallel plane. A zero Miller index would mean an infinite intercept, or an axis parallel to the plane.

Vacancies at a higher temperature

Given: $f_1 = 2.29 \times 10^{-5}$ at 400°C ; $Q = 0.76$ eV per vacancy; $k = 8.62 \times 10^{-5}$ eV/K. Find f_2 at 660°C .

1 / Use the Arrhenius relation with its prefactor

The Week 2a Bonding slides (slide 36) include a pre-exponential constant. Write the vacancy fraction as $f(T) = A \exp[-Q/(kT)]$. Assuming A and Q are unchanged between these temperatures, divide the two equations to eliminate A .

$$\begin{aligned} f_2 / f_1 &= \exp[(Q/k)(1/T_1 - 1/T_2)] \\ f_2 &= f_1 \exp[(Q/k)(1/T_1 - 1/T_2)] \end{aligned}$$

2 / Convert both temperatures to kelvin

$$\begin{aligned} T_1 &= 400 + 273.15 = 673.15 \text{ K} \\ T_2 &= 660 + 273.15 = 933.15 \text{ K} \end{aligned}$$

3 / Substitute and calculate

$$\begin{aligned} Q/k &= 0.76 / (8.62 \times 10^{-5}) = 8816.71 \text{ K} \\ (Q/k)(1/T_1 - 1/T_2) &= 3.64936 \\ f_2 &= (2.29 \times 10^{-5}) \exp(3.64936) \\ f_2 &= (2.29 \times 10^{-5})(38.4499) \\ \mathbf{f_2} &= \mathbf{8.81 \times 10^{-4} = 0.0881\%} \end{aligned}$$

This is approximately one vacant site per 1140 lattice sites. The fraction increases by about 38.45 times as temperature rises, as expected. To two significant figures, the fraction is 8.8×10^{-4} .

Why the common shortcut gives a different answer

Using $f = \exp[-Q/(kT)]$ alone sets $A = 1$. That gives 7.88×10^{-5} at 660°C , but it also predicts only 2.05×10^{-6} at 400°C , which does not match the supplied starting fraction. The given starting fraction instead implies $A \approx 11.17$. Use the ratio method to respect both the assignment data and the lecture model.

Checks: the exponent is dimensionless because Q and k use matching eV units; kelvin is mandatory; $1/T_1 - 1/T_2$ is positive; and a fraction must be multiplied by 100 to express it as a percentage.

Time needed for carbon diffusion

Given: $C_s = 1.0$ wt%, $C_0 = 0.2$ wt%, target $C = 0.6$ wt%, depth $x = 1$ mm = 1.00×10^{-3} m, and $D = 2.98 \times 10^{-11}$ m²/s at 1000°C.

1 / Choose the constant-surface diffusion solution

Assume one-dimensional diffusion into a semi-infinite solid, initially uniform carbon content, constant surface concentration, and constant D at the stated temperature.

$$(C - C_0) / (C_s - C_0) = 1 - \text{erf}(z)$$

$$z = x / [2 \sqrt{Dt}]$$

The concentration ratio is $(0.6 - 0.2)/(1.0 - 0.2) = 0.5$. Therefore $1 - \text{erf}(z) = 0.5$ and **$\text{erf}(z) = 0.5$** . Using all concentrations in wt% is valid because their units cancel.

2 / Find z using an error-function table

z	$\text{erf}(z)$
0.47	0.4937
0.48	0.5028

$$z \approx 0.47 + [(0.5000 - 0.4937)/(0.5028 - 0.4937)](0.01)$$

$$z \approx 0.47692$$

These are standard error-function values, not a normal-distribution cumulative table. A more precise inverse error function gives $z = 0.476936$; both produce the same rounded time.

3 / Rearrange for time and substitute

$$z = x / [2 \sqrt{Dt}] \text{ implies } t = x^2 / (4Dz^2)$$

$$t = (1.00 \times 10^{-3})^2 / [4(2.98 \times 10^{-11})(0.47692)^2]$$

$$t \approx 3.69 \times 10^4 \text{ s} \approx 10.2 \text{ h}$$

Check: m²/(m²/s) gives seconds. Back-substitution gives $C = 0.2 + (1.0 - 0.2)(1 - 0.5) = 0.6$ wt%. At the same target concentration, doubling the depth would require four times as long.

Avoid: setting $z = 0.5$, forgetting to square z , or using 1 mm with D in m²/s. D is already given at 1000°C, so no additional temperature correction is needed.

Source basis: supplied ENGR 290_Assign_1_2026.docx; Week 2a Bonding_2026.pptx, slide 36. Calculations independently evaluated; error-function entries rounded to four decimals.